

Chapter 2 Homework Assignment

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Problem 1: Transformations and Standardization

Let X be a continuous random variable.

1. Prove $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Sol.

$$\begin{aligned}\text{Var}(aX + b) &= E[(aX + b)^2] - E[(aX + b)]^2 \\&= E[a^2 X^2 + 2abX + b^2] - [aE(X) + b]^2 \\&= a^2 E[X^2] + 2abE[X] + b^2 - (a^2 E[X]^2 + 2abE[X] + b^2) \\&= a^2 (E[X^2] - E[X]^2) \\&= a^2 \text{Var}(X)\end{aligned}$$

2. Let $Y = \frac{X - \mu}{\sigma}$. Show that $E[Y] = 0$ and $\text{Var}(Y) = 1$

Sol.

$$\text{Let } \mu = E[X], \sigma = \sqrt{\text{Var}(X)} > 0$$

$$E[Y] = E\left[\frac{X - \mu}{\sigma}\right] = \frac{1}{\sigma}(E[X] - \mu) = \frac{1}{\sigma}(\mu - \mu) = 0$$

$$\text{Var}(Y) = \text{Var}\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \left(\frac{1}{\sigma}\right)^2 \text{Var}(X - \mu) = \frac{1}{\sigma^2} \text{Var}(X) = \frac{1}{\sigma^2} \sigma^2 = 1$$

3. If $X \sim \text{Exp}(\lambda)$ derive the CDF and verify the memoryless property.

Sol.

$$\text{CDF: } F_X(x) = P(X \leq x) = \int_0^x \lambda e^{-\lambda t} dt = [-e^{-\lambda t}]_0^x = 1 - e^{-\lambda x}$$

Thus

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0 \end{cases}$$

Let $s, t \geq 0$. Then

$$P(X > s + t \mid X > s) = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$$

Hence the exponential distribution is memoryless.

Problem 2: Normal Sampling Theory

Problem 3: Discrete Structure and Limits

Problem 4: Mixtures and Overdispersion